



Performance of a Network Link (Desempenho de uma Ligação de Rede)

Modelação e Desempenho de Redes e Serviços

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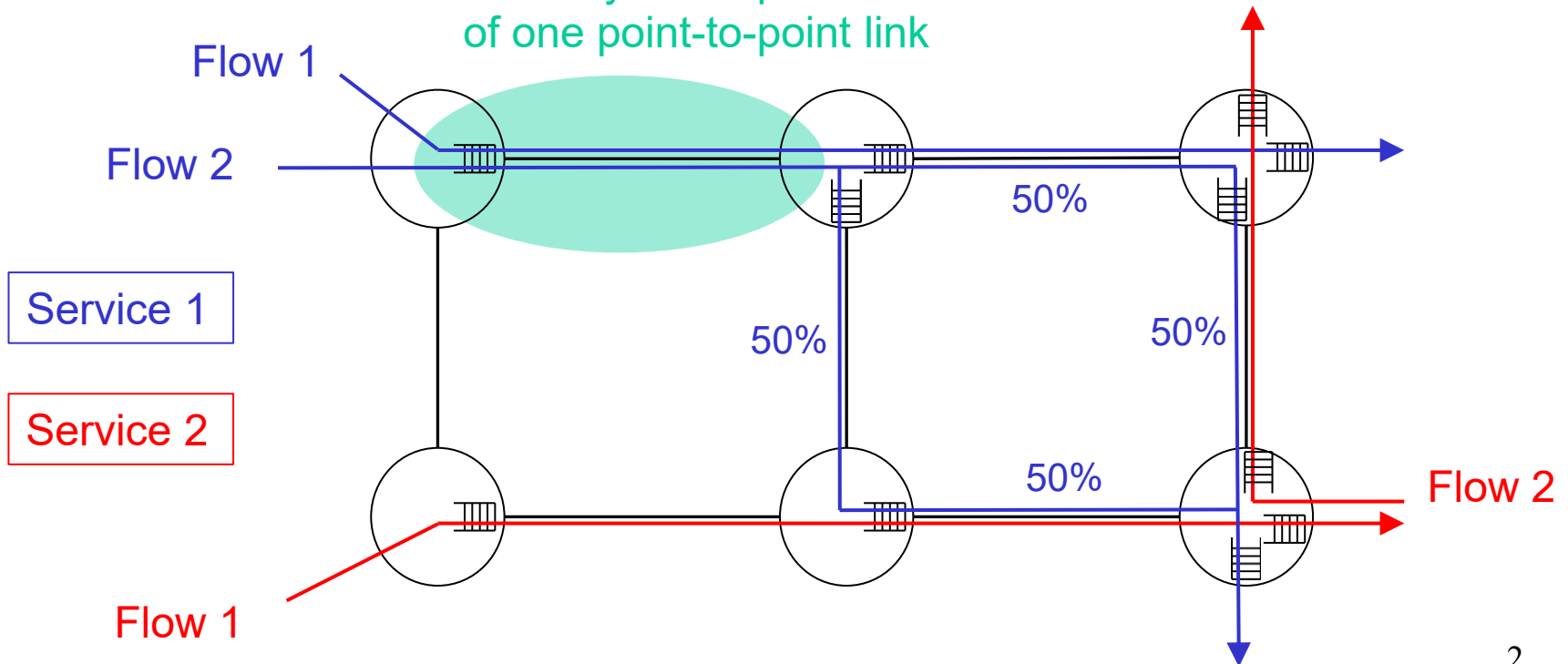
Routing in packet-switched network

A network is modelled by a set of nodes (representing routers) and a set of existing links connecting nodes.

Routing defines the sequence of links through which the packets of each flow are forwarded from the source node to the destination node.

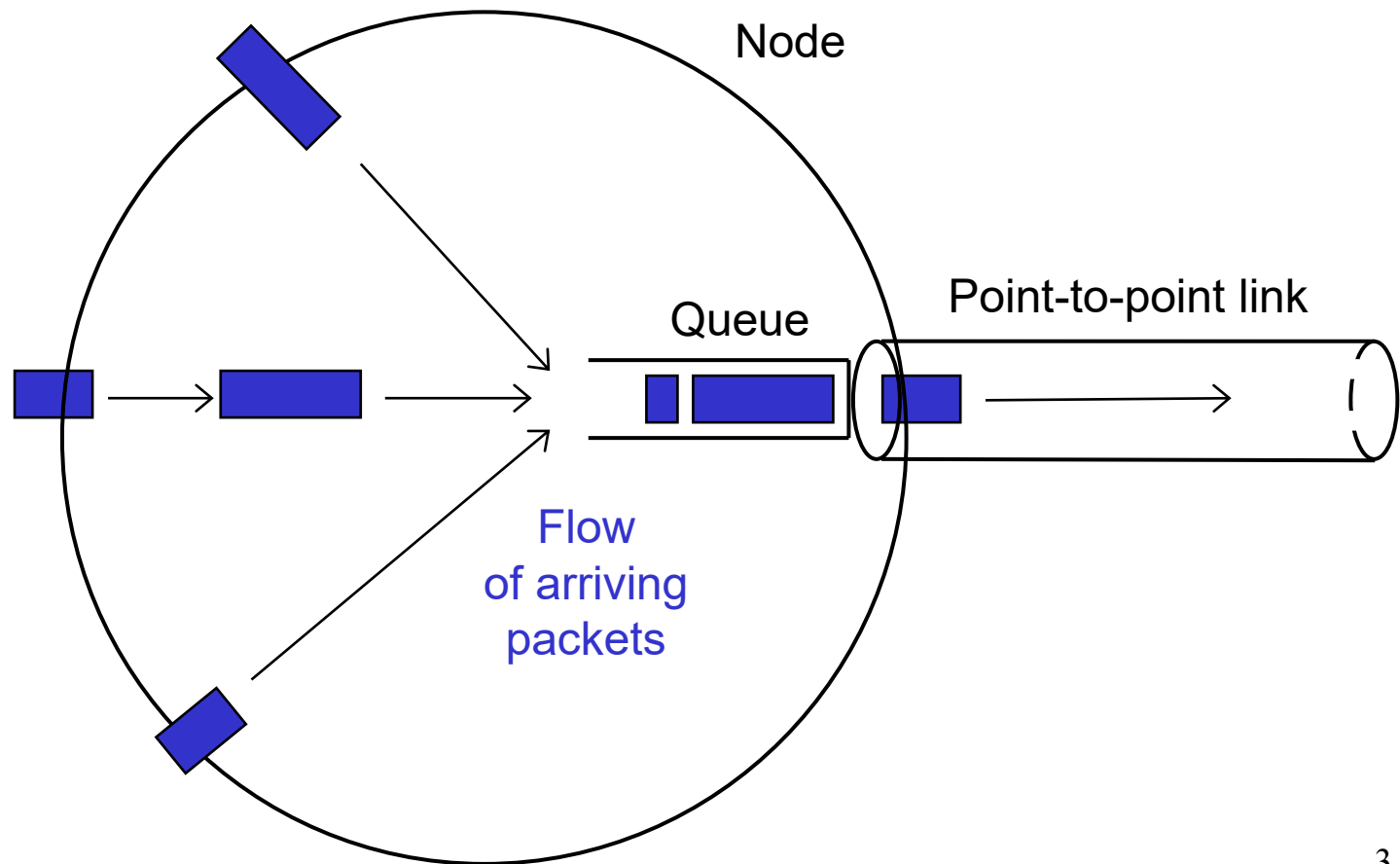
Example:

Let us analyse the performance
of one point-to-point link

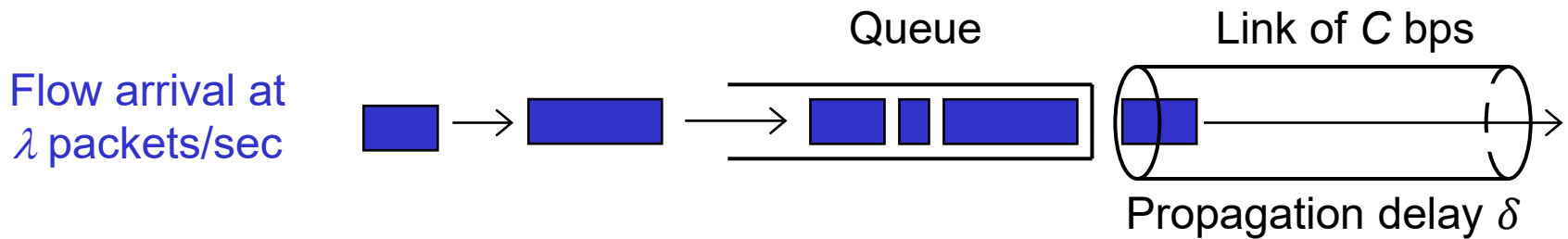


Point-to-point link

- A point-to-point link can be analysed as a server whose clients are the incoming packets and whose service is the transmission of each packet from source of the link to the destination of the link

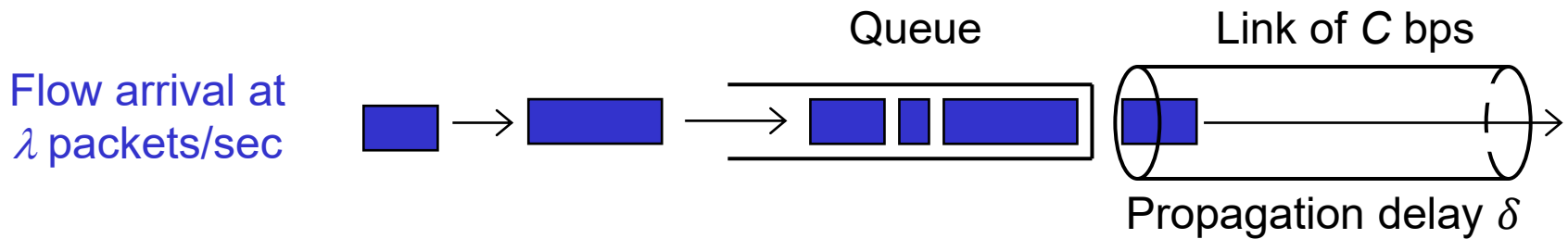


Point-to-point link



- Each arriving packet suffers a total delay on the link system which is the sum of 3 delay components:
 - the queuing delay, i.e., the time the packet stays in the queue before starting to be transmitted,
 - the transmission delay, i.e. the time the link takes to transmit all bits of the packet,
 - the propagation delay, i.e., the propagation time through the link of the last bit of the packet (the packet is fully delivered only when all its bits reach the destination of the link).

Point-to-point link



- The capacity of the link C (in bits/sec) is the speed at which the bits of each packet are transmitted through the link.
- So, the transmission time of a packet of size B bytes is the time taken by the link to transmit $8 \times B$ bits at the speed of C bits/second. For example, the transmission time of a packet of 600 Bytes on a link with capacity $C = 10$ Mbps (Megabits/sec) is:

$$S_{600} = (600 \times 8) / 10^7 = 4.8 \times 10^{-5} \text{ sec}$$

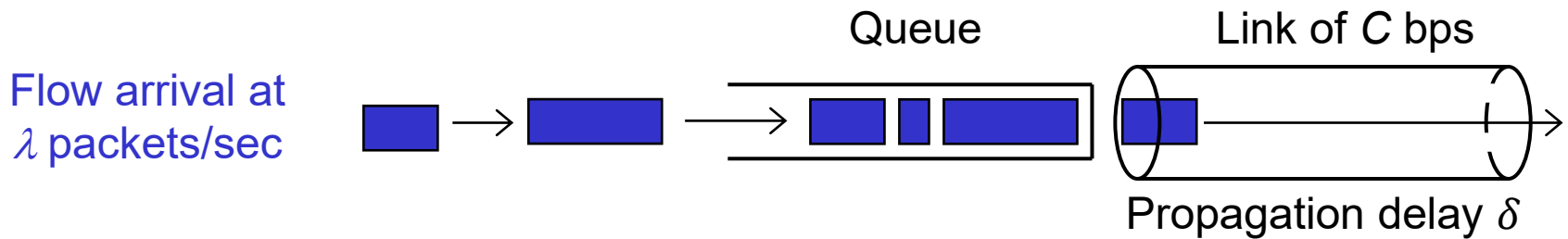
- If the average size of the arriving packets is B Bytes, the **link capacity** μ in packets/sec is the inverse of the average transmission time:

$$\mu = C / (8 \times B) \text{ pps}$$

- The **link throughput** is the average bits/sec of the transmitted packets. If packets arrive at a rate λ pps with an average packet size B , and all packets are transmitted, the link throughput is:

$$\lambda \times (8 \times B) \text{ bps}$$

Point-to-point link modelled by a $M/G/1$ system



- If the packet arrivals are a Poisson process with arriving rate λ pps (packets/sec), then, the link can be modelled by a $M/G/1$ queuing system:
 - the average queuing delay of the packets, W_Q , is given by the Pollaczek - Khintchine formula:

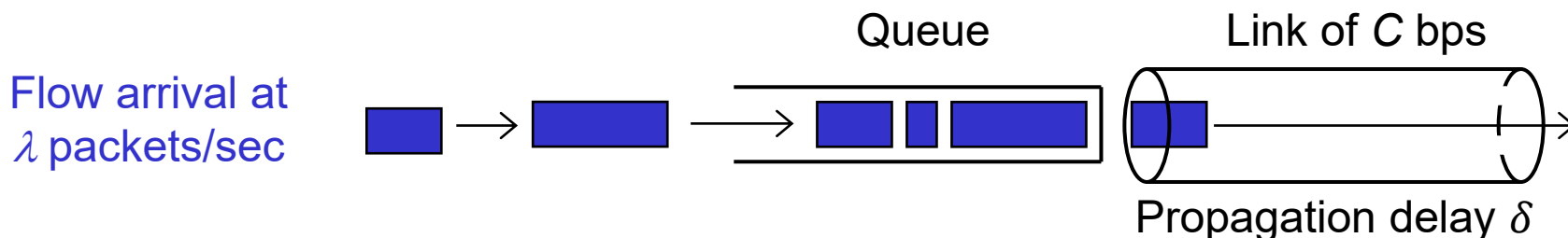
$$W_Q = \frac{\lambda E[S^2]}{2(1 - \lambda E[S])} \quad \text{valid only if } \lambda E[S] < 1$$

where S is the random variable representing the packet transmission time, $E[S]$ is its average and $E[S^2]$ is its 2nd moment

- consequently, the average system delay of the packets, W , is:

$$W = \frac{\lambda E[S^2]}{2(1 - \lambda E[S])} + E[S] + \delta$$

Point-to-point link modelled by a $M/G/1$ system



Little's theorem states that the long-term average number L of clients in a system is equal to the long-term average rate λ the clients enter the system multiplied by the average time W that each client stays in the system:

$$L = \lambda W$$

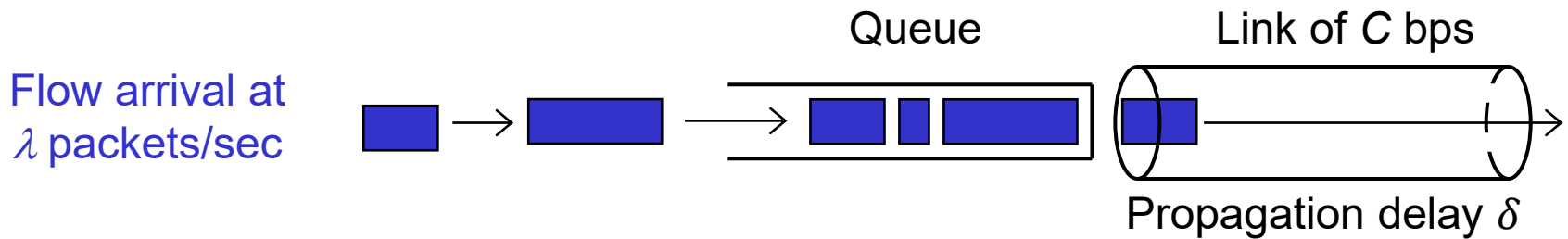
- Applying Little's theorem to the link system:
 - the average number of packets in the queue, L_Q , is:

$$L_Q = \lambda W_Q = \frac{\lambda^2 E[S^2]}{2(1 - \lambda E[S])}$$

- the average number of packets in the system, L , excluding propagation is:

$$L = \lambda W = \frac{\lambda^2 E[S^2]}{2(1 - \lambda E[S])} + \lambda E[S]$$

Point-to-point link modelled by a *M/M/1* system

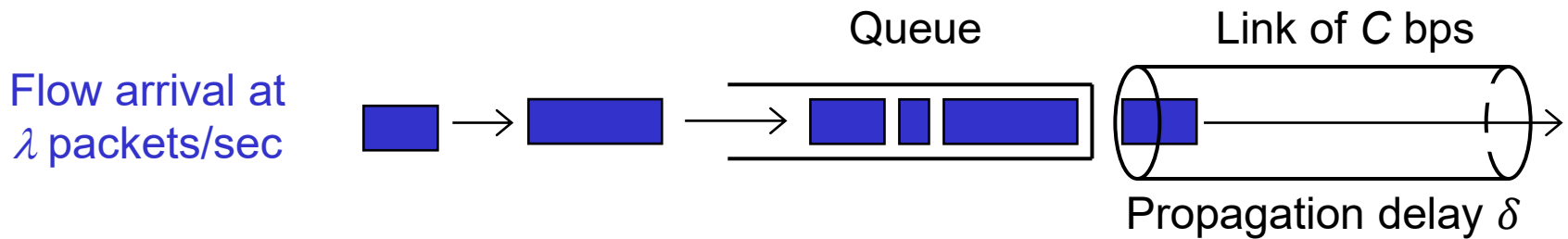


- Assuming that the packet sizes of the flow are exponentially distributed, then, the packet transmission time S is also exponentially distributed with rate μ (the capacity of the link in packets/sec).
- In this case, the link can be modelled by a *M/M/1* queuing system. Recall (from Module 1) that in a random variable with exponential distribution:

$$E[S] = \frac{1}{\mu} \quad \text{Var}[S] = \left(\frac{1}{\mu}\right)^2$$

$$\begin{aligned} \text{Var}[S] &= E[S^2] - (E[S])^2 \Leftrightarrow \\ \Leftrightarrow E[S^2] &= \text{Var}[S] + (E[S])^2 = \left(\frac{1}{\mu}\right)^2 + \left(\frac{1}{\mu}\right)^2 = 2\left(\frac{1}{\mu}\right)^2 \end{aligned}$$

Point-to-point link modelled by a *M/M/1* system



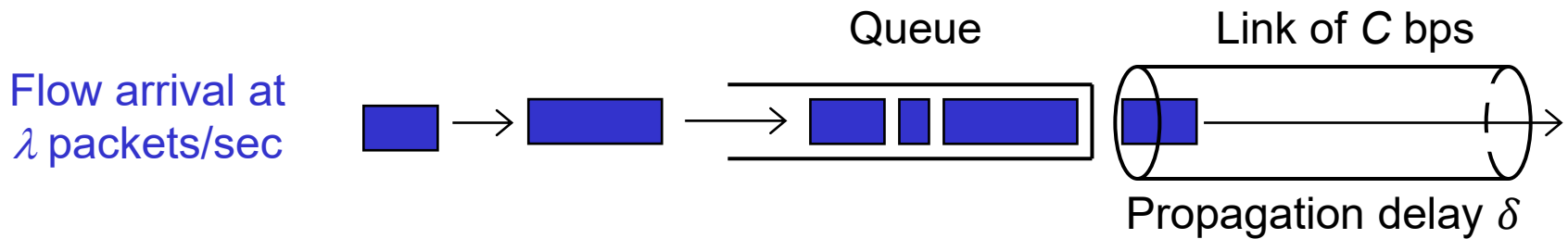
- Replacing the values of the exponential distribution in the *M/G/1* expressions, we obtain for the *M/M/1* system:
 - average queuing delay of the packets:

$$W_Q = \frac{\lambda E[S^2]}{2(1 - \lambda E[S])} = \frac{\lambda}{\mu(\mu - \lambda)}$$

- average system delay of the packets:

$$W = W_Q + E[S] + \delta = \frac{1}{(\mu - \lambda)} + \delta$$

Point-to-point link modelled by a *M/M/1* system



- Replacing the values of the exponential distribution in the *M/G/1* expressions, we obtain for the *M/M/1* system :
 - average number of packets in the queue:

$$L_Q = \lambda W_Q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

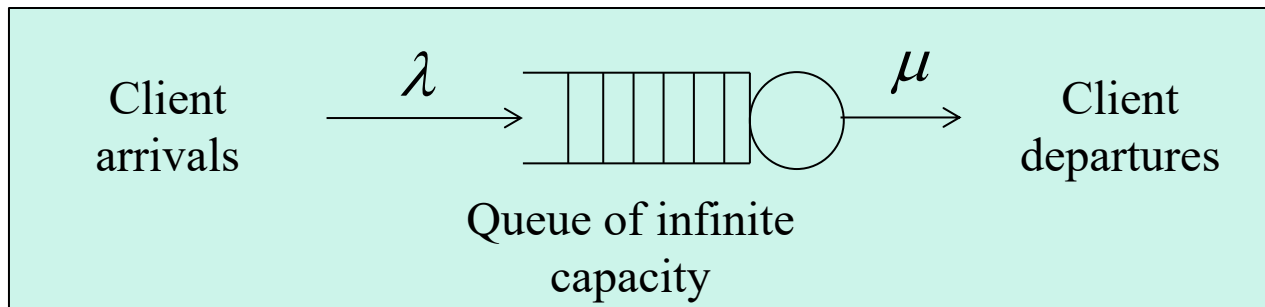
- average number of packets in the system, excluding propagation:

$$L = \lambda W = \frac{\lambda}{(\mu - \lambda)}$$

***M/M/1* queuing system**

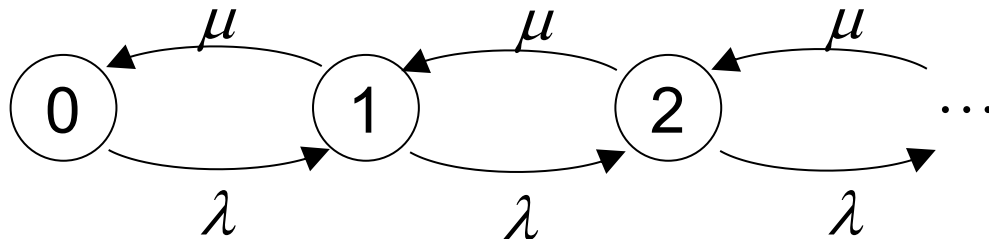
A *M/M/1* queuing system is a CTMC (Continuous Time Markov Chain) of type birth-death:

1. client arrivals are a Poisson process with arriving rate λ
2. the system has 1 server
3. the serving time of each client is exponentially distributed with average $1/\mu$ (i.e., the serving rate of the server is μ)
4. the queue is of infinite size
5. queued clients are served with a first-in-first-out policy



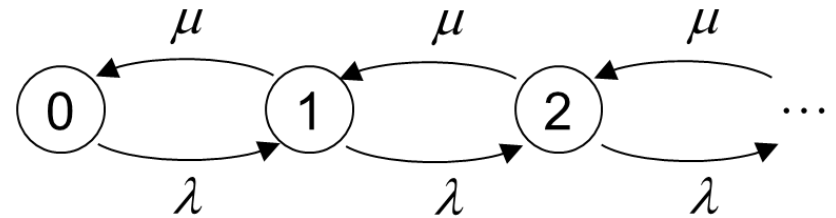
***M/M/1* queuing system**

- The state transition diagram of a *M/M/1* system has an infinite number of states and is given by:



- Each state n (with $n = 0, 1, 2, \dots, +\infty$) represents the system being with n clients
 - state n (with $n \geq 1$) represents the system with 1 client being served and $n - 1$ clients being in the queue
- The birth rates are all equal to λ as the client arrivals are independent of the system state
- The death rates are all equal to μ as there is only one server attending clients, whatever number of clients are in the system

M/M/1 system (limit state probabilities)



$$\pi_0 = \frac{1}{1 + \sum_{i=1}^{\infty} \left(\frac{\lambda_0 \lambda_1 \dots \lambda_{i-1}}{\mu_1 \mu_2 \dots \mu_i} \right)} = \frac{1}{1 + \sum_{i=1}^{\infty} \left(\frac{\lambda}{\mu} \right)^i}$$

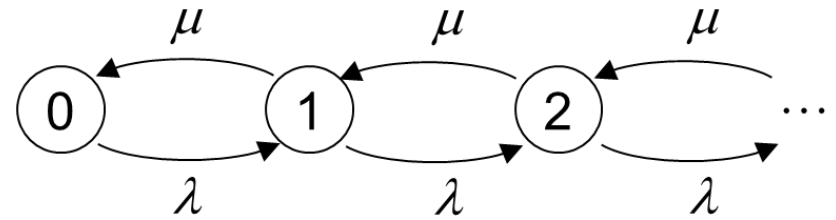
$$\pi_n = \frac{\lambda_0 \lambda_1 \dots \lambda_{n-1}}{\mu_1 \mu_2 \dots \mu_n} \pi_0 = \left(\frac{\lambda}{\mu} \right)^n \pi_0 = \frac{\left(\frac{\lambda}{\mu} \right)^n}{1 + \sum_{i=1}^{\infty} \left(\frac{\lambda}{\mu} \right)^i}, n = 1, 2, \dots$$

Equivalently:

$$\pi_n = \frac{\left(\frac{\lambda}{\mu} \right)^n}{\sum_{i=0}^{\infty} \left(\frac{\lambda}{\mu} \right)^i}, n = 0, 1, 2, \dots$$

Necessary condition for the existence of the limit probabilities: $\frac{\lambda}{\mu} < 1$

M/M/1 system (limit state probabilities)



From the general expression:

$$\pi_n = \frac{\left(\frac{\lambda}{\mu}\right)^n}{\sum_{i=0}^{\infty} \left(\frac{\lambda}{\mu}\right)^i}, n = 0, 1, 2, \dots$$

we can deduce that the probability of the system having n clients is:

$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right), n = 0, 1, 2, \dots$$

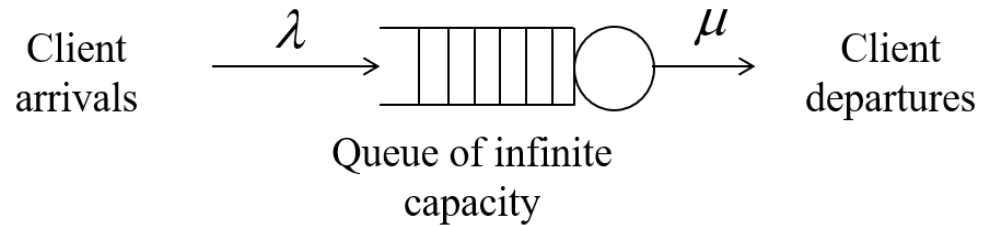
Using:

$$\frac{1}{1-x} = x^0 + x + x^2 + x^3 \dots, 0 < x < 1$$

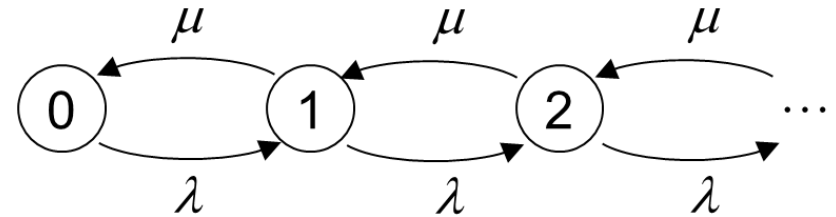
and the probability of the system having n or more clients is:

$$\sum_{i=n}^{\infty} \pi_i = \left(\frac{\lambda}{\mu}\right)^n, n = 0, 1, 2, \dots$$

M/M/1 system (performance parameters)



$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right), n = 0, 1, 2, \dots$$



- Average number of clients in the system:

$$L = \sum_{n=1}^{\infty} n \times \pi_n = \frac{\lambda}{\mu - \lambda}$$

Using:

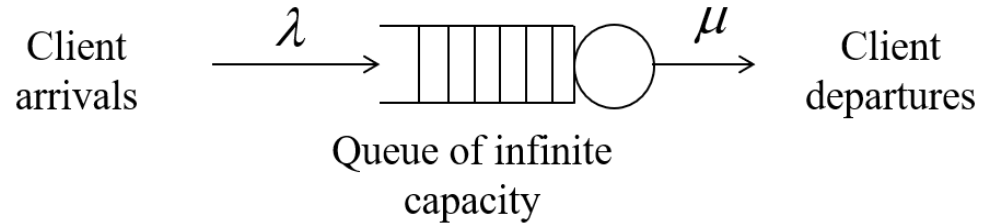
$$\frac{1}{1-x} = x^0 + x + x^2 + x^3 \dots, 0 < x < 1$$

- Average time each client stays in the system:

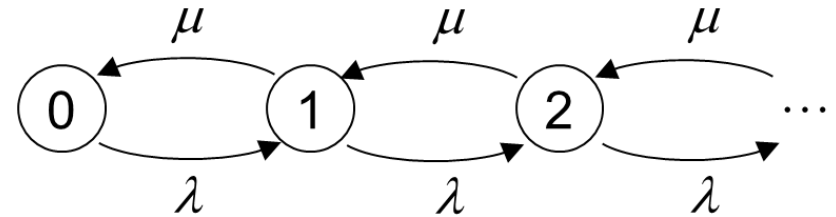
$$L = \lambda W \quad \rightarrow \quad W = \frac{L}{\lambda} = \frac{1}{\mu - \lambda}$$

Little's theorem

M/M/1 system (performance parameters)



$$\pi_n = \left(\frac{\lambda}{\mu}\right)^n \left(1 - \frac{\lambda}{\mu}\right), n = 0, 1, 2, \dots$$



- Average time each client waits in the queue:

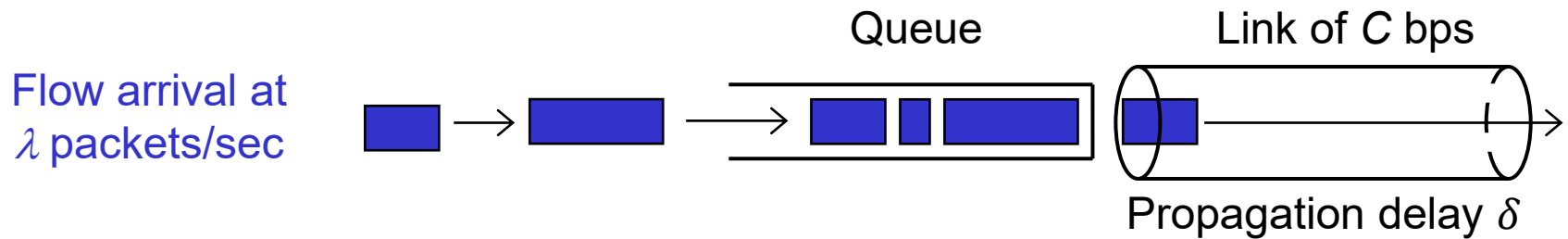
$$W_Q = W - \frac{1}{\mu} = \frac{1}{(\mu - \lambda)} - \frac{1}{\mu} = \frac{\lambda}{\mu(\mu - \lambda)}$$

- Average number of clients in the queue:

$$L_Q = \lambda W_Q \quad \rightarrow \quad L_Q = \frac{\lambda^2}{\mu(\mu - \lambda)}$$

Little's theorem

Point-to-point link with finite queue capacity

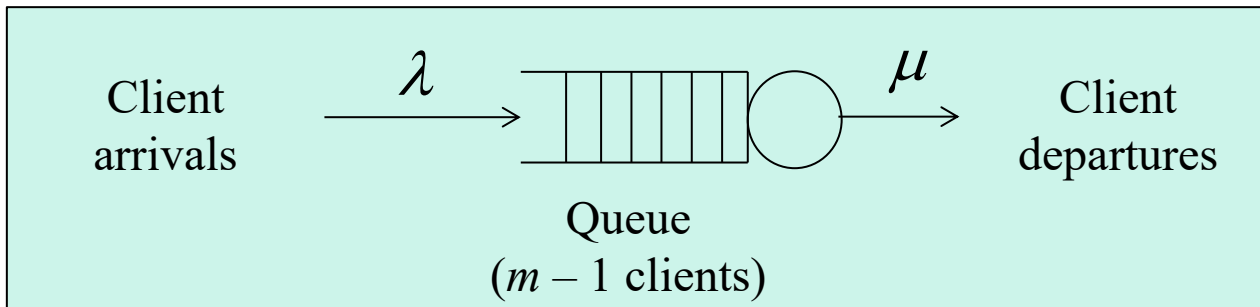


- When the packet sizes are assumed to be exponentially distributed, the use of a CTMC of type birth-dead (as it is the $M/M/1$ system) allows further system analysis as it provides the probabilistic characterization of the system occupancy.
- This is particularly important when queue capacity (in maximum number of queued packets) is not infinite. In this case, the link can be modelled by a $M/M/1/m$ queuing system, which is also a CTMC of type birth-dead (next slides).

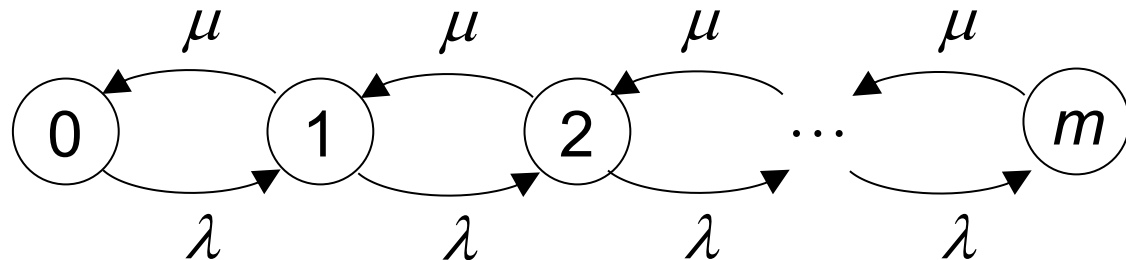
***M/M/1/m* system**

A *M/M/1/m* queuing system is a CTMC of type birth-death:

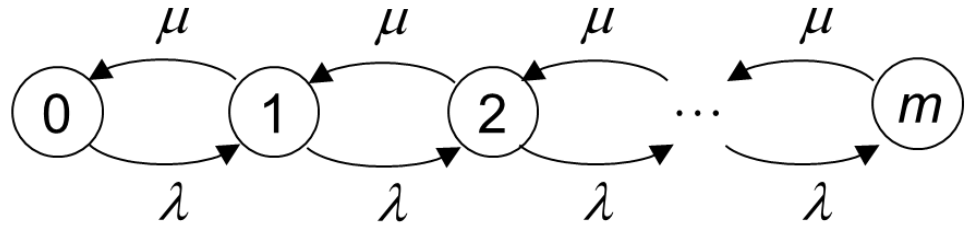
1. client arrivals are a Poisson process with arriving rate λ
2. the system has 1 server
3. the serving time of each client is exponentially distributed with average $1/\mu$ (i.e., the serving rate of the server is μ)
4. the system has a capacity of m clients (1 being served and $m - 1$ being in the queue)
5. queued clients are served with a first-in-first-out policy



State transition
diagram



***M/M/1/m* system**



- The probability of n clients in the system is:

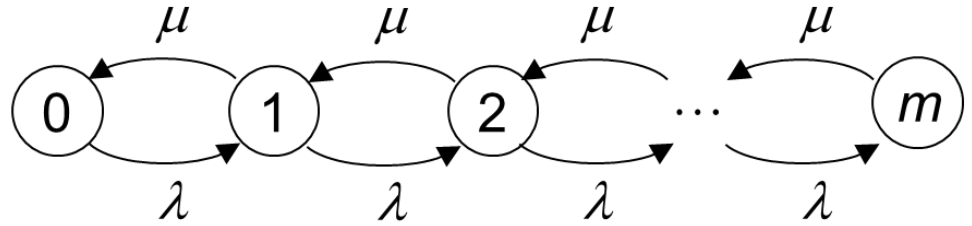
$$\pi_n = \frac{\left(\frac{\lambda}{\mu}\right)^n}{\sum_{i=0}^m \left(\frac{\lambda}{\mu}\right)^i}, \quad n = 0, 1, 2, \dots, m$$

PASTA property: clients arriving according to a Poisson process such that the serving time of each client is statistically independent of its arrival time instance see time averages (i.e., the proportion of arriving customers who find the system in each state is equal to the long-run proportion of time the system is in the state).

- Therefore, the probability of an arrival client being rejected (i.e., the server not available and the queue fully occupied) is given by the probability of state m :

$$\pi_m = \frac{\left(\frac{\lambda}{\mu}\right)^m}{\sum_{i=0}^m \left(\frac{\lambda}{\mu}\right)^i}$$

***M/M/1/m* system**



From the general expression:
$$\pi_n = \frac{\left(\frac{\lambda}{\mu}\right)^n}{\sum_{i=0}^m \left(\frac{\lambda}{\mu}\right)^i}, n = 0, 1, 2, \dots, m$$

and considering $\rho = \frac{\lambda}{\mu} \neq 1$, we can deduce that the state probabilities are given by:

$$\pi_n = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^n, n = 0, 1, 2, \dots, m$$

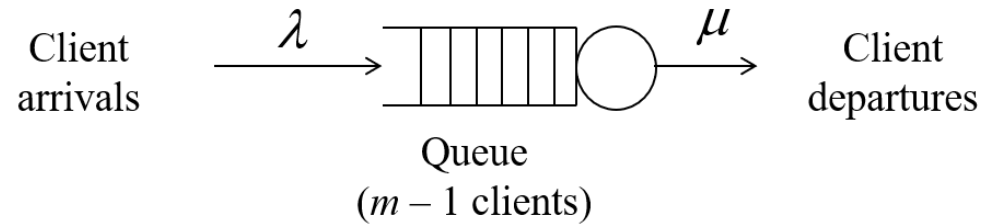
and the probability of an arrival client being rejected is:

$$\pi_m = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^m$$

If $\rho = 1$ (i.e., $\lambda = \mu$) then, all states have the same probability:

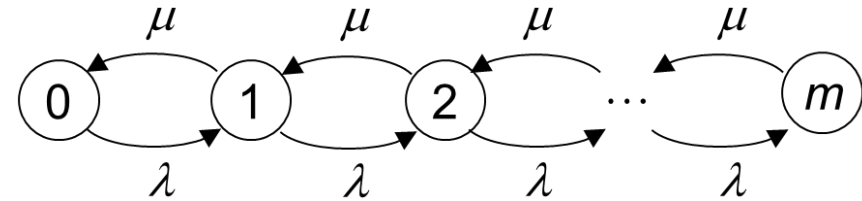
$$\pi_n = \frac{1}{m + 1}, n = 0, 1, 2, \dots, m$$

M/M/1/m system (performance parameters)



$$\pi_n = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^n, n = 0, 1, 2, \dots, m$$

$$\rho = \frac{\lambda}{\mu}$$



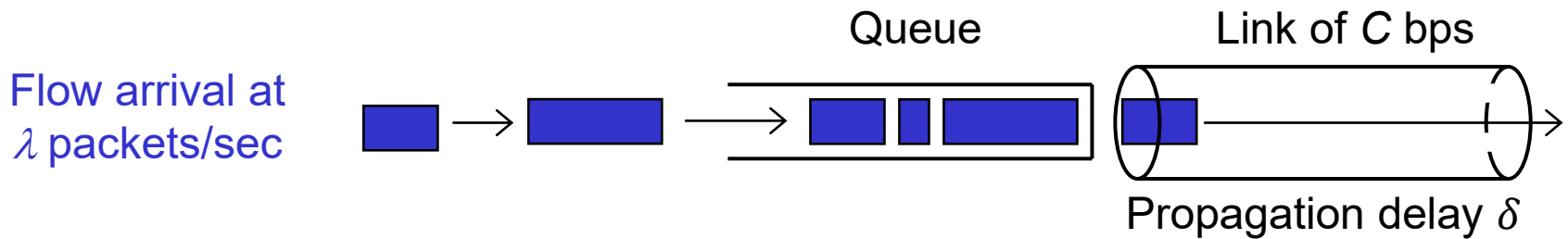
- Client rejection probability: $\pi_m = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^m$
- Average number of clients in the system: $L = \sum_{n=0}^m n \times \pi_n$
- Average time each non-rejected client stays in the system:

$$L = \lambda W \quad \rightarrow \quad W = \frac{L}{\lambda(1 - \pi_m)}$$

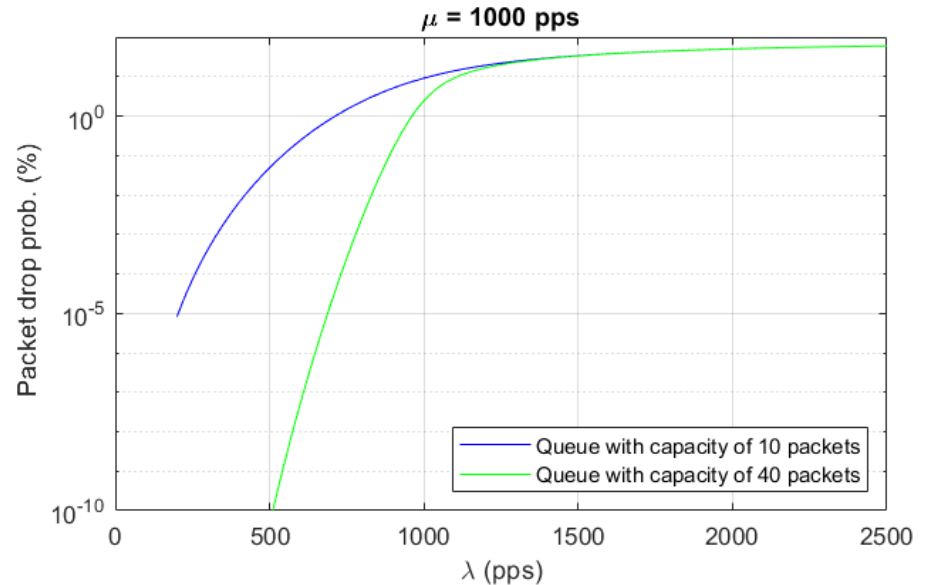
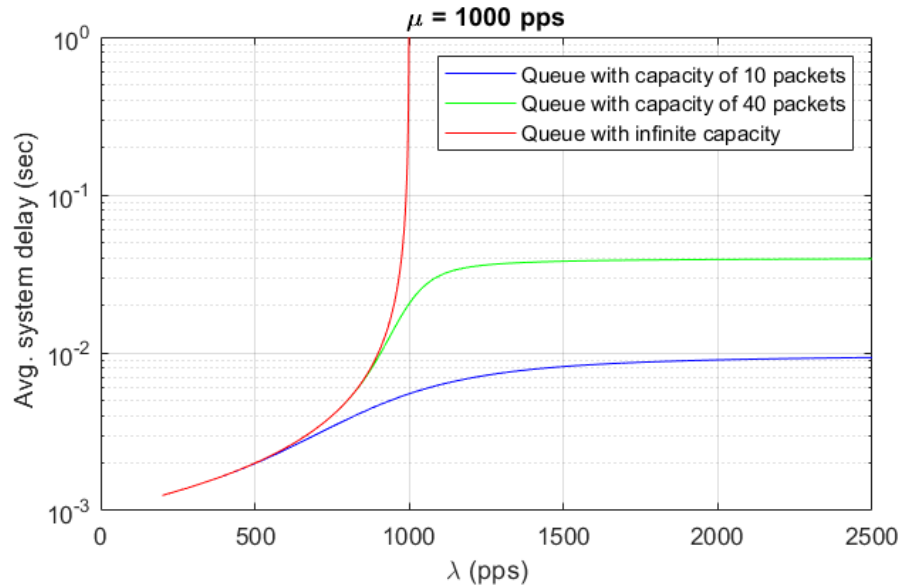
Little's theorem

- Average time each non-rejected client waits in the queue: $W_Q = W - \frac{1}{\mu}$

Impact of different queue capacities



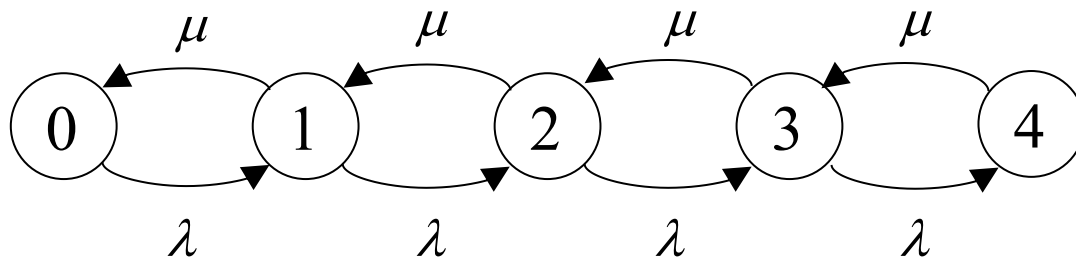
- If the queue has an infinite capacity, the system is modelled by an $M/M/1$
- In the queue has a capacity of $m-1$ packets, the system is modelled by an $M/M/1/m$
- Example: Link of 10 Mbps, average packet size of 1250 Bytes, $\delta = 0$
 $\mu = 10^7/(1250 \times 8) = 1000$ pps



Example 1

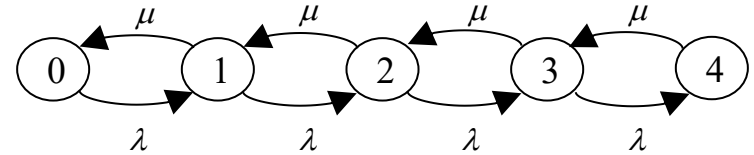
Consider a data packet transmission system with a queue of capacity 3 packets and a link of 1 Mbps. The link is supporting a data flow of packets whose arrivals are a Poisson process with rate 200 pps (packets/sec). The packets length is exponentially distributed with average 500 bytes. Determine:

- (a) the packet loss probability,
- (b) the percentage of packets not suffering queuing delay,
- (c) the average packet delay of packets accepted by the system.



M/M/1/4 model

Example 1 – solution of (a)



Consider a data packet transmission system with a queue of capacity 3 packets and a link of 1 Mbps. The link is supporting a data flow of packets whose arrivals are a Poisson process with rate 200 pps (packets/sec). The packets length is exponentially distributed with average 500 bytes. Determine:

(a) the packet loss probability,

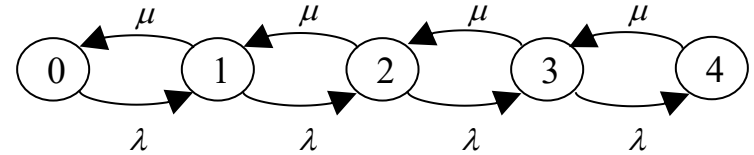
$$\mu = \frac{10^6 \text{ bps}}{500 \times 8 \text{ bpp}} = 250 \text{ pps} \quad \lambda = 200 \text{ pps}$$

$$\pi_4 = \frac{\left(\frac{\lambda}{\mu}\right)^4}{\left(\frac{\lambda}{\mu}\right)^0 + \left(\frac{\lambda}{\mu}\right)^1 + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 + \left(\frac{\lambda}{\mu}\right)^4}$$

$$\pi_4 = \frac{\left(\frac{200}{250}\right)^4}{1 + \frac{200}{250} + \left(\frac{200}{250}\right)^2 + \left(\frac{200}{250}\right)^3 + \left(\frac{200}{250}\right)^4} = 0.1218 = 12.18\%$$

$$\pi_n = \frac{\left(\frac{\lambda}{\mu}\right)^n}{\sum_{i=0}^m \left(\frac{\lambda}{\mu}\right)^i}, \quad n = 0, 1, 2, \dots, m$$

Example 1 – solution of (b)



Consider a data packet transmission system with a queue of capacity 3 packets and a link of 1 Mbps. The link is supporting a data flow of packets whose arrivals are a Poisson process with rate 200 pps (packets/sec). The packets length is exponentially distributed with average 500 bytes. Determine:

(b) the percentage of packets not suffering queuing delay,

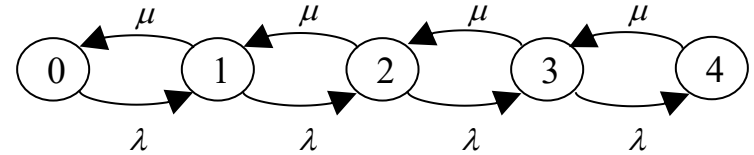
$$\mu = \frac{10^6 \text{ bps}}{500 \times 8 \text{ bpp}} = 250 \text{ pps} \quad \lambda = 200 \text{ pps}$$

$$\pi_0 = \frac{\left(\frac{\lambda}{\mu}\right)^0}{\left(\frac{\lambda}{\mu}\right)^0 + \left(\frac{\lambda}{\mu}\right)^1 + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 + \left(\frac{\lambda}{\mu}\right)^4}$$

$$\pi_0 = \frac{1}{1 + \frac{200}{250} + \left(\frac{200}{250}\right)^2 + \left(\frac{200}{250}\right)^3 + \left(\frac{200}{250}\right)^4} = 0.2975 = 29.75\%$$

$$\pi_n = \frac{\left(\frac{\lambda}{\mu}\right)^n}{\sum_{i=0}^m \left(\frac{\lambda}{\mu}\right)^i}, \quad n = 0, 1, 2, \dots, m$$

Example 1 – solution of (c)



Consider a data packet transmission system with a queue of capacity 3 packets and a link of 1 Mbps. The link is supporting a data flow of packets whose arrivals are a Poisson process with rate 200 pps (packets/sec). The packets length is exponentially distributed with average 500 bytes. Determine:

(c) the average packet delay of packets accepted by the system.

$$\mu = \frac{10^6 \text{ bps}}{500 \times 8 \text{ bpp}} = 250 \text{ pps} \quad \lambda = 200 \text{ pps}$$

$$\pi_n = \frac{\left(\frac{\lambda}{\mu}\right)^n}{\sum_{i=0}^m \left(\frac{\lambda}{\mu}\right)^i}, n = 0, 1, 2, \dots, m$$

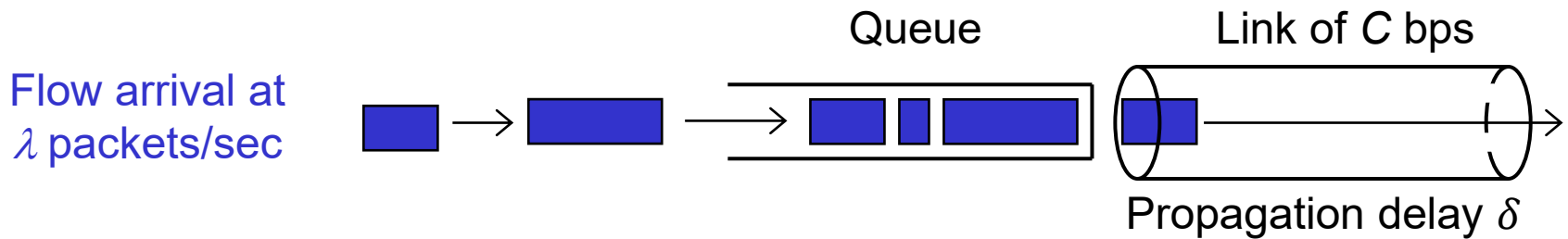
$$L = \frac{\left(\frac{\lambda}{\mu}\right)^1 + 2 \times \left(\frac{\lambda}{\mu}\right)^2 + 3 \times \left(\frac{\lambda}{\mu}\right)^3 + 4 \times \left(\frac{\lambda}{\mu}\right)^4}{\left(\frac{\lambda}{\mu}\right)^0 + \left(\frac{\lambda}{\mu}\right)^1 + \left(\frac{\lambda}{\mu}\right)^2 + \left(\frac{\lambda}{\mu}\right)^3 + \left(\frac{\lambda}{\mu}\right)^4} = 1.5631 \text{ packets}$$

$$L = \sum_{n=0}^m n \times \pi_n$$

$$W = \frac{L}{\lambda(1 - \pi_m)}$$

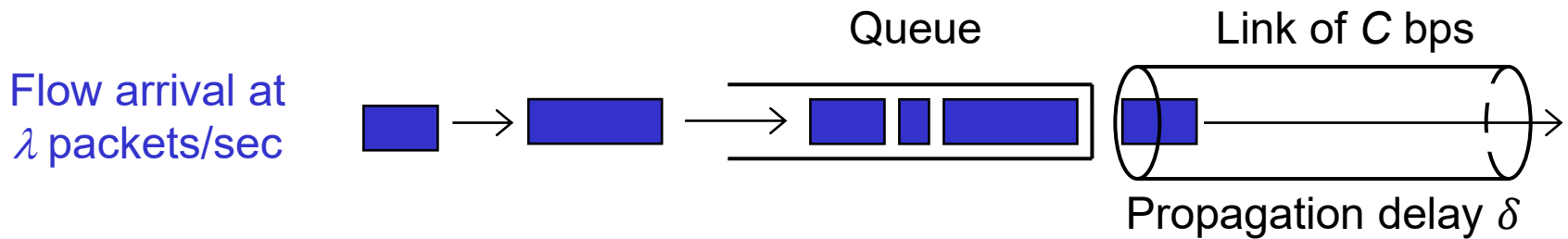
$$W = \frac{L}{\lambda(1 - \pi_m)} = \frac{1.5631}{200 \times (1 - 0.1218)} = 0.0089 \text{ sec}$$

Point-to-point link with a bit error rate



- As seen before, packet loss can happen when the queue has a finite capacity (arriving packets are discarded when they arrive and the queue is full)
- If the link has a non-residual BER (bit error rate), packet loss can also happen due to propagation errors in the link (packets are discarded at the destination node of the link when they arrive with at least one errored bit)
- Packet loss due to BER has a more negative impact on the performance:
 - In packet loss due to queue fullness, packets are discarded at the entry of the system. Therefore, they do not occupy system resources (queue and transmission link).
 - In packet loss due to BER, packets are discarded at the exit of the system. Therefore, they occupy system resources and penalise the delay suffered by the packets that are successfully transmitted without errors

Point-to-point link with a bit error rate



- The packet loss due to BER can be modelled using a binomial distribution and depends on the packet size (as seen in Module 1 of the course unit).
- We can estimate the packet loss probability p_{BER} due to a BER value b using the average size B_{avg} (in Bytes) of the flow packets:

$$p_{BER} = 1 - (1 - b)^{8 \times B_{avg}}$$

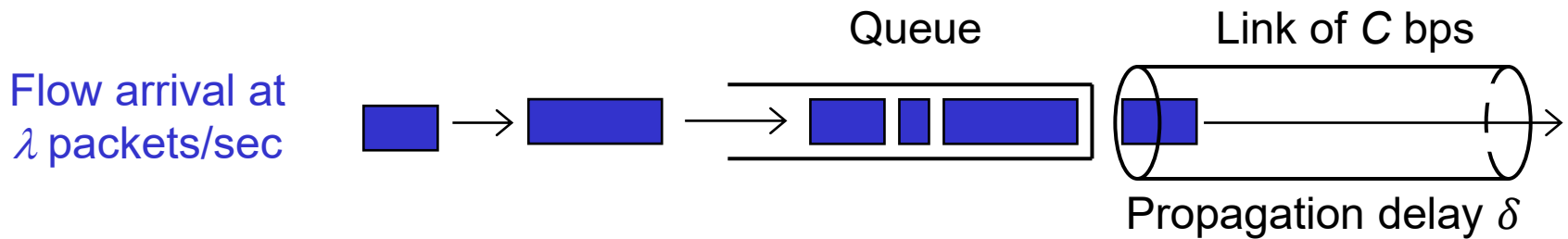
- In a system with both a finite queue and a link with BER, where p_Q is the packet loss probability due to queue fullness, the total packet loss probability p_L becomes:

$$p_L = p_Q + (1 - p_Q) \times p_{BER}$$

Probability of packets
discard at the entry

Probability of packets being transmitted
and then discarded at the exit

Point-to-point link with a bit error rate



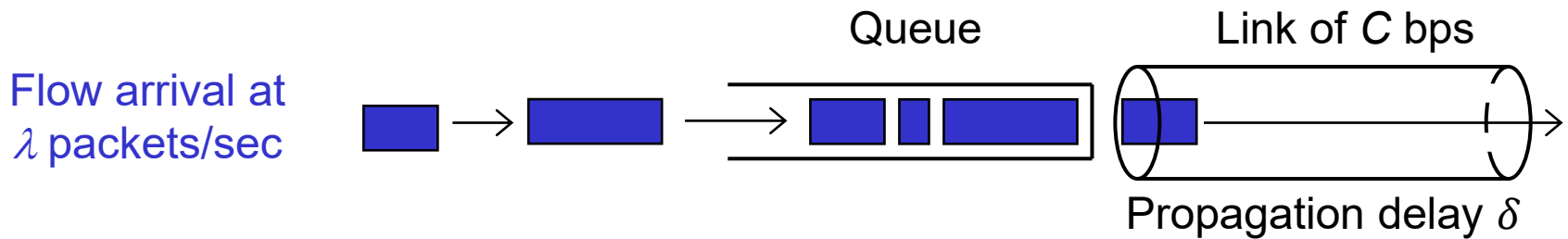
- The **link throughput** is the rate (in bits/sec) at which the packets are transmitted by the link. Since all packets accepted in the system are transmitted, the link throughput is:

$$\lambda \times (1 - p_Q) \times (8 \times B_{avg}) \text{ bps}$$

- The **link goodput** is similar to the link throughput but it accounts only for the transmitted packets that are successfully received (i.e., received without errors). So, the link goodput is:

$$\lambda \times (1 - p_Q) \times (1 - p_{BER}) \times (8 \times B_{avg}) \text{ bps}$$

Point-to-point link with a bit error rate



- The exact determination of packet loss probability, link throughput and link goodput requires the knowledge of the packet size mass probability function.
- If packet sizes are $B_1, B_2, \dots, B_N$ (in Bytes), with probabilities $p_1, p_2, \dots, p_N$, respectively, first, the average packet size is:

$$B_{avg} = \sum_{i=1}^N p_i \times B_i$$

and the link throughput is: $\lambda \times (1 - p_Q) \times (8 \times B_{avg})$ bps

- Then, the packet loss probability due to a BER value b becomes:

$$p_{BER} = \sum_{i=1}^N p_i \times (1 - (1 - b)^{8 \times B_i})$$

and the link goodput becomes:

$$\lambda \times (1 - p_Q) \times \sum_{i=1}^N (p_i \times (1 - b)^{8 \times B_i} \times (8 \times B_i)) \text{ bps}$$

Example 2

Consider a data packet transmission system with a very large queue and a link of $C = 1$ Mbps. The link introduces a BER of 10^{-5} and is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 200$ pps. The packets size is 200 Bytes with probability 40% or 600 Bytes with probability 60%. Determine:

- (a) the packet loss probability,
- (b) the link throughput,
- (c) the link goodput,
- (d) the average packet delay.

Example 2 – solution of (a)

Consider a data packet transmission system with a very large queue and a link of $C = 1$ Mbps. The link introduces a BER of 10^{-5} and is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 200$ pps. The packets size is 200 Bytes with probability 40% or 600 Bytes with probability 60%. Determine:

(a) the packet loss probability,

Since the queue is very large: $p_Q = 0$

$$p_L = p_Q + (1 - p_Q) \times p_{BER}$$

$$p_{BER} = \sum_{i=1}^N p_i \times (1 - (1 - b)^{8 \times B_i})$$

$$\begin{aligned} p_L = p_{BER} &= p_1 \times (1 - (1 - b)^{8 \times B_1}) + p_2 \times (1 - (1 - b)^{8 \times B_2}) = \\ &= 0.4 \times (1 - (1 - 10^{-5})^{8 \times 200}) + 0.6 \times (1 - (1 - 10^{-5})^{8 \times 600}) = \\ &= 0.4 \times 0.015873 + 0.6 \times 0.04686 = \\ &= 0.0345 \end{aligned}$$

Example 2 – solution of (b)

Consider a data packet transmission system with a very large queue and a link of $C = 1$ Mbps. The link introduces a BER of 10^{-5} and is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 200$ pps. The packets size is 200 Bytes with probability 40% or 600 Bytes with probability 60%. Determine:

(b) the link throughput,

Since the queue is very large: $p_Q = 0$

$$B_{avg} = \sum_{i=1}^N p_i \times B_i$$

$$\begin{aligned} &\text{Link throughput (in bps):} \\ &\lambda \times (1 - p_Q) \times (8 \times B_{avg}) \end{aligned}$$

$$B_{avg} = p_1 \times B_1 + p_2 \times B_2 = 0.4 \times 200 + 0.6 \times 600 = 440 \text{ Bytes}$$

Link throughput =

$$\lambda \times (8 \times B_{avg}) = 200 \times 8 \times 440 = 704000 \text{ bps} = 0.704 \text{ Mbps}$$

Example 2 – solution of (c)

Consider a data packet transmission system with a very large queue and a link of $C = 1$ Mbps. The link introduces a BER of 10^{-5} and is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 200$ pps. The packets size is 200 Bytes with probability 40% or 600 Bytes with probability 60%. Determine:

(c) the link goodput,

Since the queue is very large: $p_Q = 0$

$$\text{Link goodput (in bps):}$$
$$\lambda \times (1 - p_Q) \times \sum_{i=1}^N (p_i \times (1 - b)^{8 \times B_i} \times (8 \times B_i))$$

Link goodput =

$$\begin{aligned} & \lambda \times (p_1 \times (1 - b)^{8 \times B_1} \times (8 \times B_1) + p_2 \times (1 - b)^{8 \times B_2} \times (8 \times B_2)) = \\ & = 200 \times (0.4 \times (1 - 10^{-5})^{8 \times 200} \times (8 \times 200) + 0.6 \times (1 - 10^{-5})^{8 \times 600} \times (8 \times 600)) \\ & = 674973 \text{ bps} = 0.675 \text{ Mbps} \end{aligned}$$

Recall from (b) that the
link throughput is 0.704 Mbps

Example 2 – solution of (d)

Consider a data packet transmission system with a very large queue and a link of $C = 1$ Mbps. The link introduces a BER of 10^{-5} and is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 200$ pps. The packets size is 200 Bytes with probability 40% or 600 Bytes with probability 60%. Determine:

(d) the average packet delay.

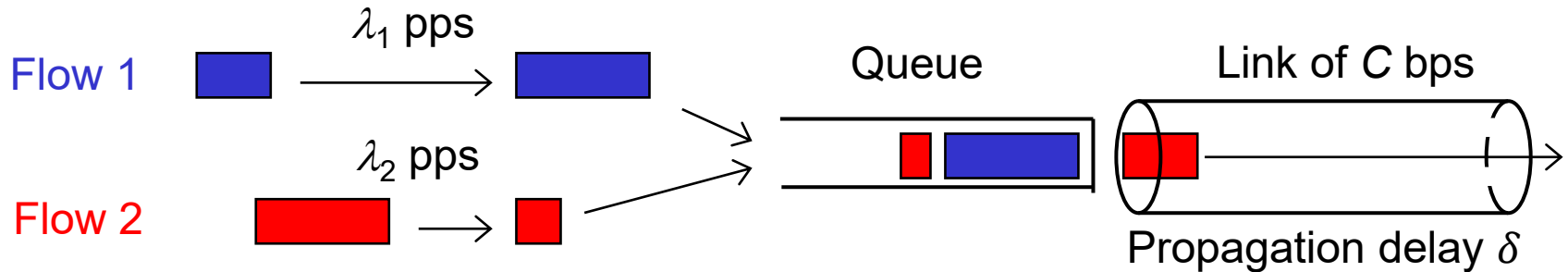
$$W = \frac{\lambda E[S^2]}{2(1 - \lambda E[S])} + E[S]$$

$$\begin{aligned} E[S] &= p_1 \times \frac{8 \times B_1}{C} + p_2 \times \frac{8 \times B_2}{C} \\ &= 0.4 \times \frac{8 \times 200}{10^6} + 0.6 \times \frac{8 \times 600}{10^6} = 2.881 \times 10^{-3} \text{ sec} \end{aligned}$$

$$\begin{aligned} E[S^2] &= p_1 \times \left(\frac{8 \times B_1}{C} \right)^2 + p_2 \times \left(\frac{8 \times B_2}{C} \right)^2 = 0.4 \times \left(\frac{8 \times 200}{10^6} \right)^2 + 0.6 \times \left(\frac{8 \times 600}{10^6} \right)^2 \\ &= 1.485 \times 10^{-5} \text{ sec}^2 \end{aligned}$$

$$\begin{aligned} W &= \frac{\lambda E[S^2]}{2(1 - \lambda E[S])} + E[S] = \frac{200 \times 1.485 \times 10^{-5}}{2 \times (1 - 200 \times 2.881 \times 10^{-3})} + 2.881 \times 10^{-3} \\ &= 8.54 \times 10^{-3} \text{ sec} = 8.54 \text{ msec} \end{aligned}$$

Point-to-point link supporting multiple services



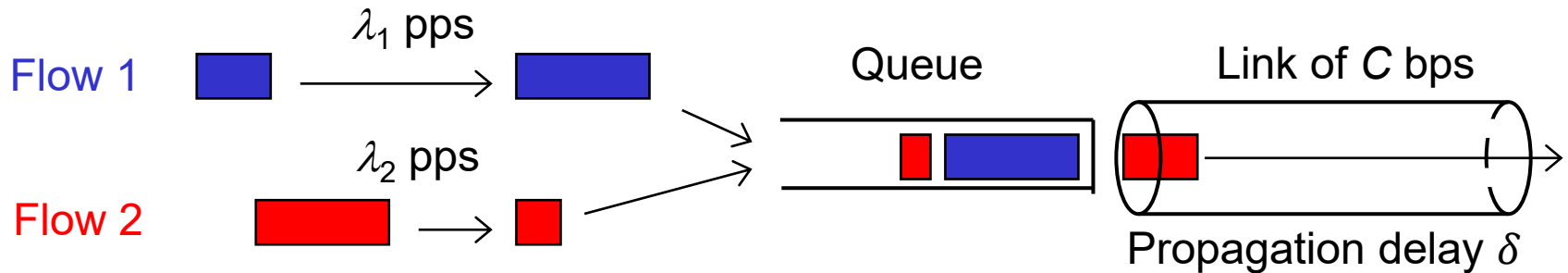
- Let us consider now that the link is supporting multiple flows of packets (possibly of different services)
- When the queuing of the packets of all flows is done in a single FIFO (first-in-first-out) queue, we say that the flows are statistically multiplexed by the link.
- If the packet arrivals of each flow is a Poisson process, the packet arrivals of the aggregate of all flows is also a Poisson process with arriving rate:

$$\lambda = \sum_{i=1}^N \lambda_i$$

where N is the number of flows and λ_i is the arriving rate of each flow i .

- In this case, whatever queuing system is the more adequate to analyse it, **the average packet queuing delay is the same for all flows**.

Point-to-point link supporting multiple services

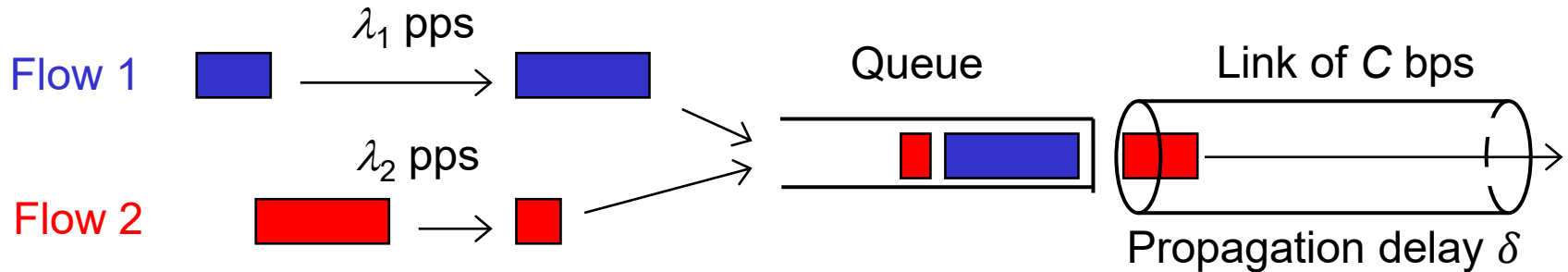


- If the queue has infinite size, the system can always be modelled as a $M/G/1$ queuing system.
- The average packet queuing delay W_Q is equal for all flows and given by the Pollaczek - Khintchine formula:

$$W_Q = \frac{\lambda E[S^2]}{2(1 - \lambda E[S])}$$

where S is the random variable representing the packet transmission time of the aggregate of all flows, $E[S]$ is its average and $E[S^2]$ is its 2nd moment.

Point-to-point link supporting multiple services



- Assuming that the packet sizes of the aggregate of all flows is exponentially distributed, the link can again be modelled by:
 - a $M/M/1$ queuing system (when the queue is of infinite size) or
 - a $M/M/1/m$ queuing model (when the queue is of finite size).
- To obtain a good approximation, first the average packet size B_{avg} of the aggregate should be determined as the weighted average of the average packet sizes of each flow:

$$B_{avg} = \frac{\sum_{i=1}^N \lambda_i \times B_i}{\sum_{i=1}^N \lambda_i} \text{ Bytes}$$

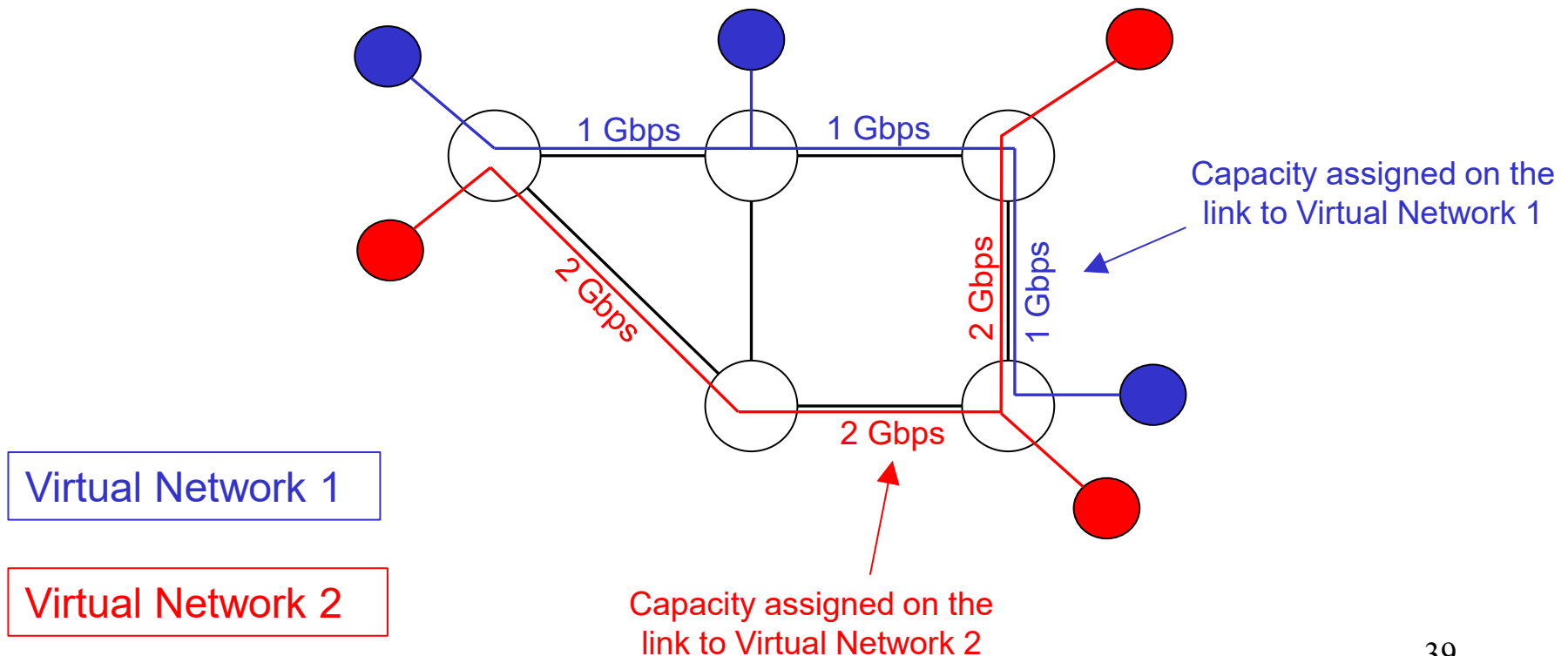
where N is the number of flows, λ_i is the arriving rate of each flow i and B_i is the average packet size of each flow i .

- Then, the link capacity is: $\mu = \frac{C}{8 \times B_{avg}} \text{ pps}$

Network virtualization

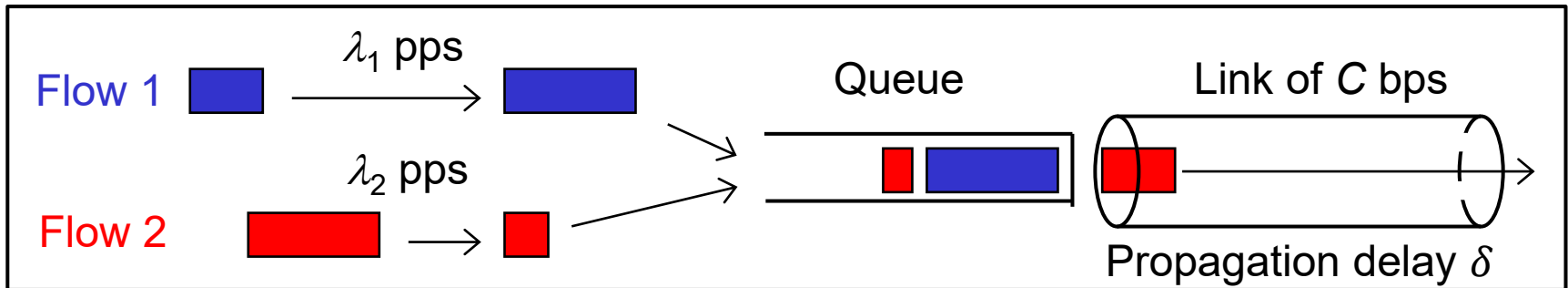
Network virtualization is the process of abstracting physical network resources and combining them into a single, software-based administrative entity called a virtual network.

Example (all physical links with a capacity of 10 Gbps each)

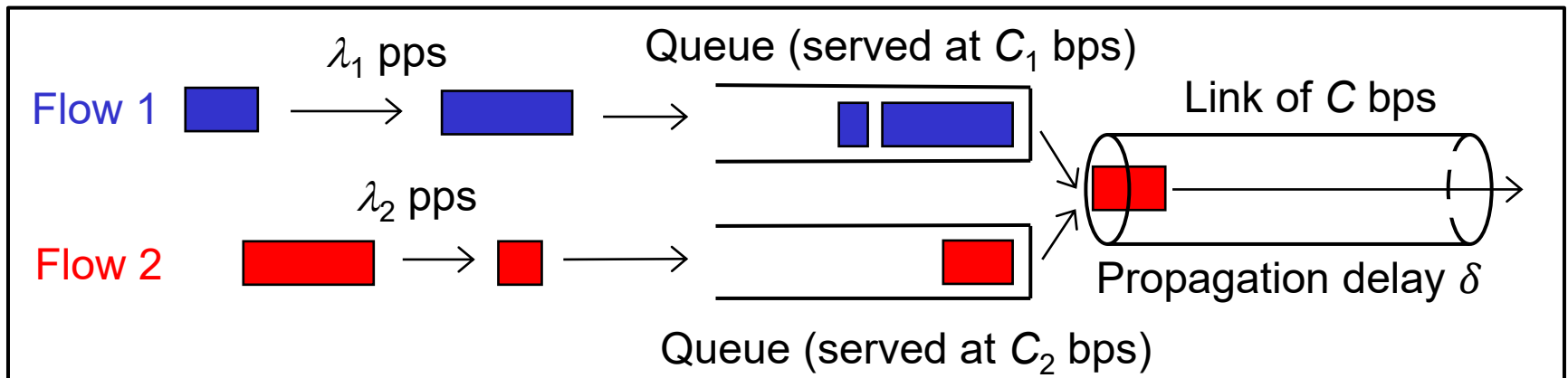


Point-to-point link with assigned resources

- Multiple flows are said to be statistically multiplexed when the queuing of the packets of all flows is done in a single FIFO queue (the previous case):

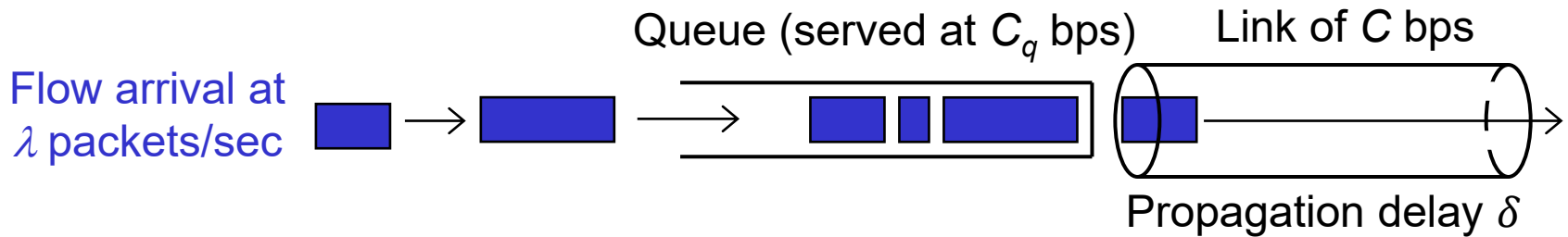


- With C_i resources assigned for each flow i :



If the link scheduler is perfect, the performance of the link for each flow is independent of the other flows and, therefore, can be analysed separately.

Point-to-point link with assigned resources



- In this case, the packets in the queue are served at a rate of C_q bps, while the packets are transmitted at a rate of C bps. Consider:

$$\alpha = \frac{C}{C_q} > 1$$

- If S represents the packet transmission time, then, $\alpha \times S$ represents the packet queuing time with the average $\alpha \times E[S]$ and 2nd moment $\alpha^2 \times E[S^2]$.
- With a queue of infinite size, the average packet queuing delay is:

$$W_Q = \frac{\lambda \alpha^2 E[S^2]}{2(1 - \lambda \alpha E[S])}$$

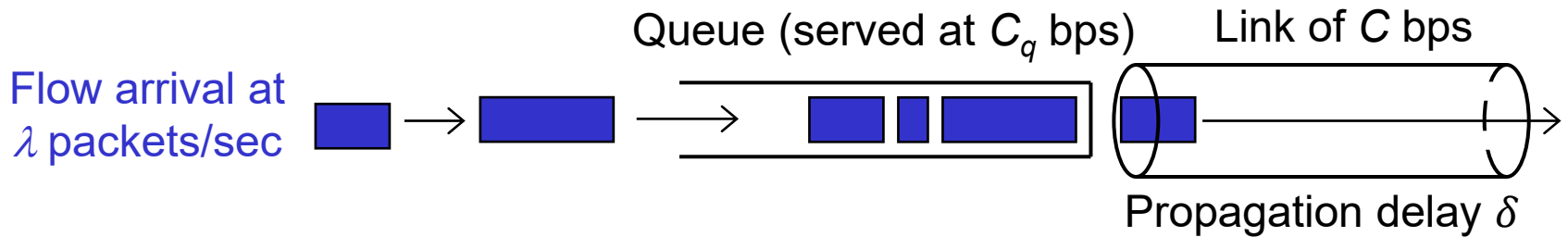
valid if $\lambda \alpha E[S] < 1$

and the average system delay of the packets is:

$$W = \frac{\lambda \alpha^2 E[S^2]}{2(1 - \lambda \alpha E[S])} + E[S] + \delta$$

average link
transmission time

Point-to-point link with assigned resources



- Assuming that the packet sizes are exponentially distributed, the queue of the link can be modelled by a $M/M/1$ system when the queue is of infinite size.
- If the average size of the arriving packets is B Bytes, the **link capacity** μ and the **queue serving capacity** μ_q are:

$$\mu = \frac{C}{8 \times B} \quad \mu_q = \frac{C_q}{8 \times B}$$

- Therefore, the average packet queuing delay W_Q and average number of packets in the queue L_Q (recall the $M/M/1$ performance formulas) are:

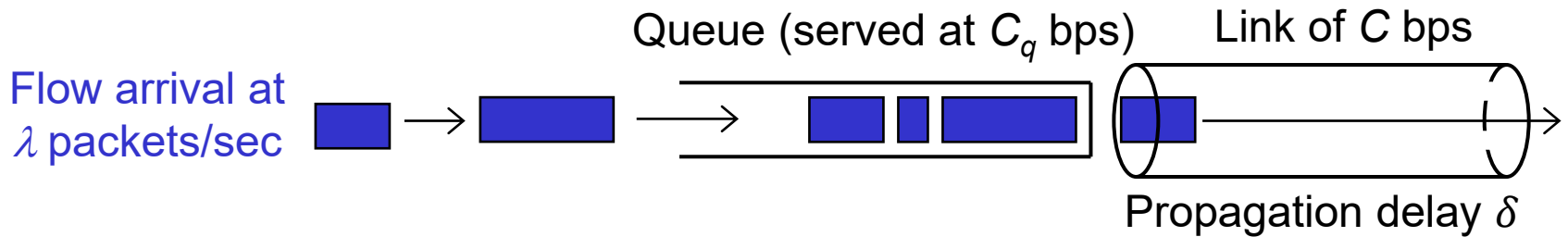
$$W_Q = \frac{\lambda}{\mu_q(\mu_q - \lambda)} \quad L_Q = \frac{\lambda^2}{\mu_q(\mu_q - \lambda)}$$

while the average system delay of the packets is:

$$W = \frac{\lambda}{\mu_q(\mu_q - \lambda)} + \frac{1}{\mu} + \delta$$

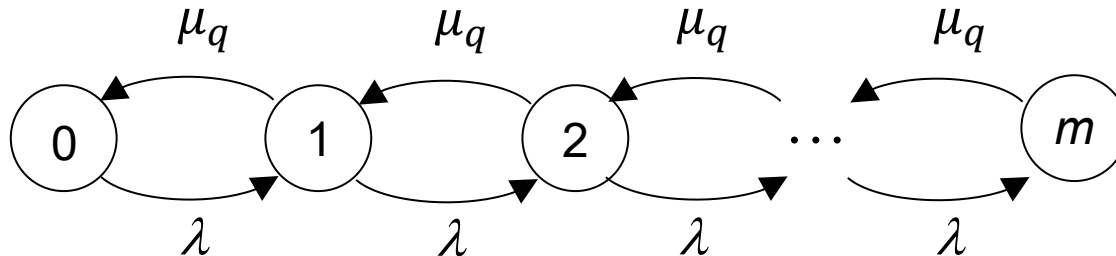
link capacity (in pps)

Point-to-point link with assigned resources



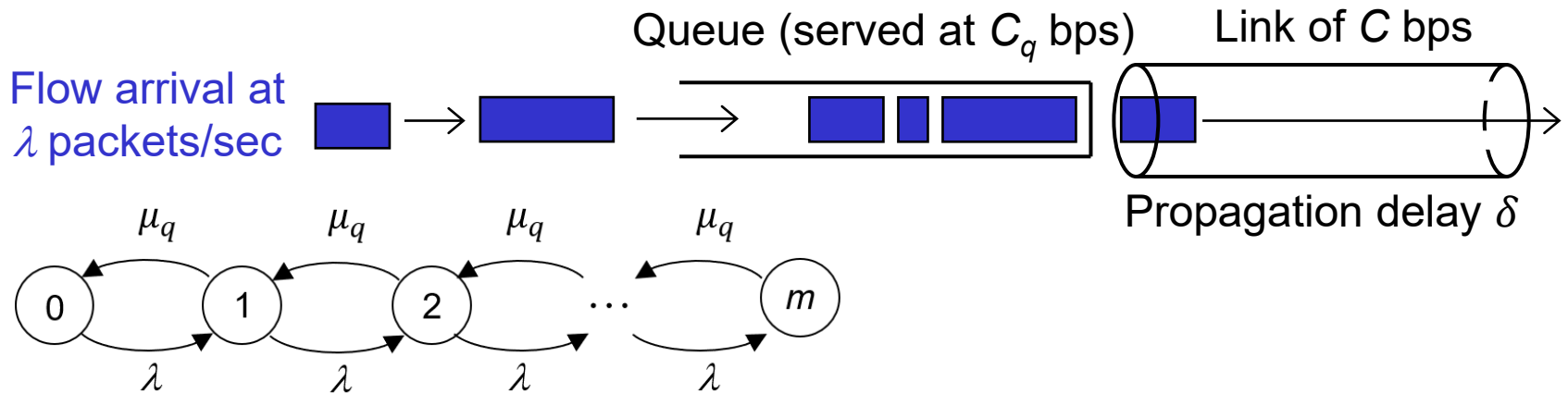
- Assuming that the packet sizes are exponentially distributed, the queue of the link can be modelled by a $M/M/1/m$ system when the queue is of size $m - 1$.
- Again, if the average size of the arriving packets is B Bytes, the **link capacity** μ and the **queue serving capacity** μ_q are:

$$\mu = \frac{C}{8 \times B} \quad \mu_q = \frac{C_r}{8 \times B}$$
- First, the state transition diagram of the birth-death process representing the queue occupation is:



where state n (with $n \geq 1$) represents the queue with $n - 1$ clients.

Point-to-point link with assigned resources



- Considering $\rho = \frac{\lambda}{\mu_q}$, the packet loss probability is: $\pi_m = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^m$

the average number of packets in the queue L_Q and the average packet queuing delay W_Q are:

$$L_Q = \sum_{n=2}^m (n-1) \times \pi_n$$

$$W_Q = \frac{L_Q}{\lambda(1 - \pi_m)}$$

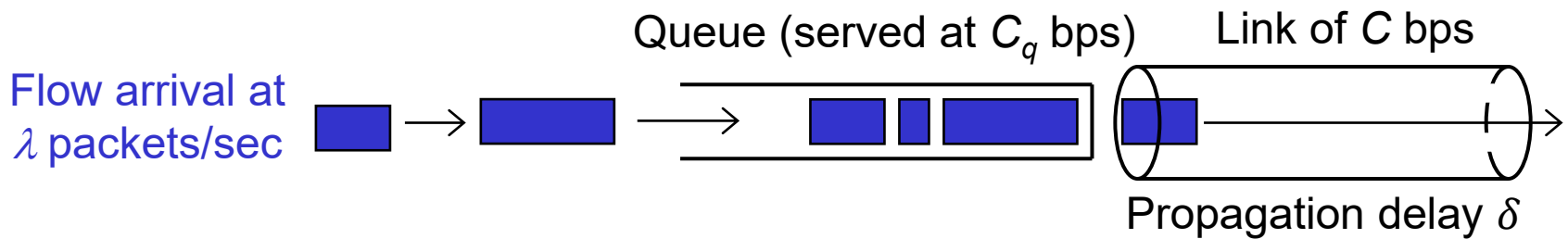
applying
Little's theorem

while the average system delay of the packets is:

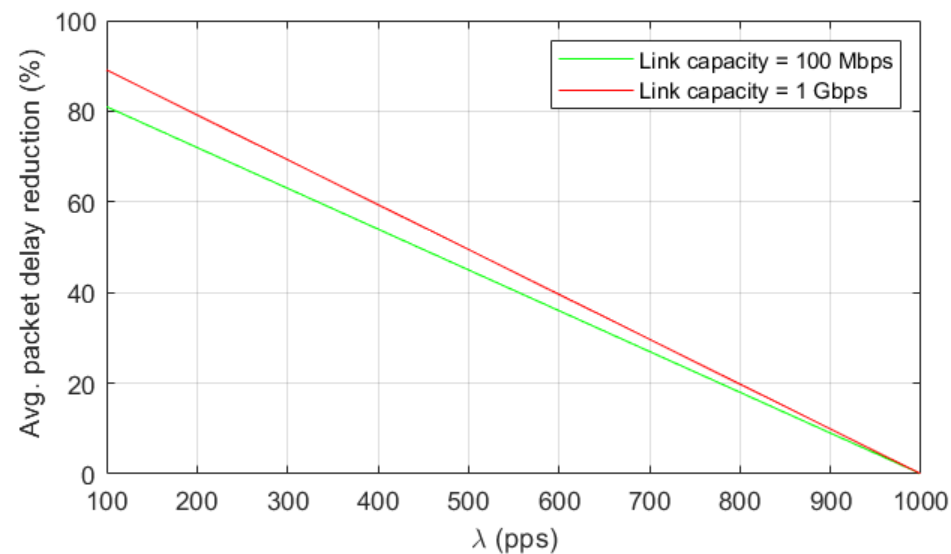
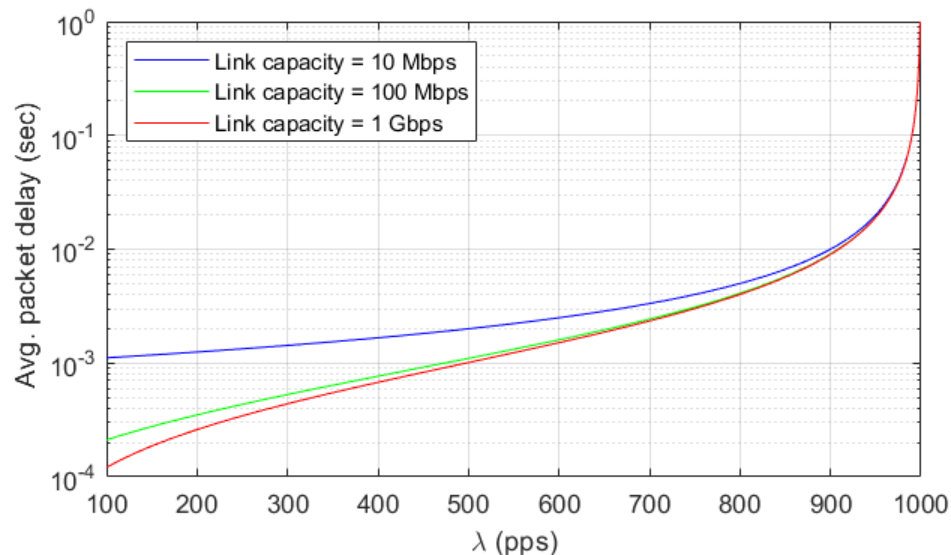
$$W = W_Q + \frac{1}{\mu} + \delta$$

link capacity (in pps)

Impact of link capacity with assigned resources



- The assignment C_q influences the average packet queuing delay while the link capacity C influences the packet transmission time.
- Example: Assignment of $C_q = 10$ Mbps, queue of infinite size, exponentially distributed packet sizes with average 1250 Bytes, $\delta = 0$
 $\mu_a = 10^7/(1250 \times 8) = 1000$ pps



Example 3

Consider a packet transmission system with a queue of size X packets and a link of capacity $C = 100$ Mbps and propagation delay $\delta = 10$ μ sec. The system is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 900$ pps and the packet sizes are exponentially distributed with average 500 Bytes. Determine the average system delay of the packets when:

- (a) the flow can use the total link capacity and X is infinite,
- (b) the flow is assigned with 10 Mbps and X is infinite,
- (c) the flow is assigned with 10 Mbps and $X = 3$ packets.

Example 3 – solution of (a)

Consider a packet transmission system with a queue of size X packets and a link of capacity $C = 100$ Mbps and propagation delay $\delta = 10$ μ sec. The system is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 900$ pps and the packet sizes are exponentially distributed with average 500 Bytes. Determine the average system delay of the packets when:

(a) the flow can use the total link capacity and X is infinite,

$M/M/1$ system:

$$W = \frac{1}{(\mu - \lambda)} + \delta$$

$$\mu = \frac{C}{8 \times B} = \frac{100 \times 10^6}{8 \times 500} = 25000 \text{ pps}$$

$$W = \frac{1}{(\mu - \lambda)} + \delta = \frac{1}{25000 - 900} + 10^{-5} = 5.15 \times 10^{-5} \text{ sec} = 51.5 \mu\text{sec}$$

Example 3 – solution of (b)

Consider a packet transmission system with a queue of size X packets and a link of capacity $C = 100$ Mbps and propagation delay $\delta = 10$ μ sec. The system is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 900$ pps and the packet sizes are exponentially distributed with average 500 Bytes. Determine the average system delay of the packets when:

(b) the flow is assigned with 10 Mbps and X is infinite,

$$\mu = \frac{C}{8 \times B} = \frac{100 \times 10^6}{8 \times 500} = 25000 \text{ pps}$$

$$W = \frac{\lambda}{\mu_q(\mu_q - \lambda)} + \frac{1}{\mu} + \delta$$

$$C_q = 10 \times 10^6 \text{ bps} \longrightarrow \mu_q = \frac{C_q}{8 \times B} = \frac{10 \times 10^6}{8 \times 500} = 2500 \text{ pps}$$

$$\begin{aligned} W &= \frac{\lambda}{\mu_q(\mu_q - \lambda)} + \frac{1}{\mu} + \delta = \frac{900}{2500 \times (2500 - 900)} + \frac{1}{25000} + 10^{-5} \\ &= 2.75 \times 10^{-4} \text{ sec} = 275.0 \mu\text{sec} \end{aligned}$$

Example 3 – solution of (c)

Consider a packet transmission system with a queue of size X packets and a link of capacity $C = 100$ Mbps and propagation delay $\delta = 10$ μ sec. The system is supporting a flow of packets whose arrivals are a Poisson process with rate $\lambda = 900$ pps and the packet sizes are exponentially distributed with average 500 Bytes. Determine the average system delay of the packets when:

(c) the flow is assigned with 10 Mbps and $X = 3$ packets.

$$C_q = 10 \times 10^6 \text{ bps}$$

$$\mu = 25000 \text{ pps}$$

$$\mu_q = 2500 \text{ pps}$$

$$\rho = \frac{\lambda}{\mu_q} = \frac{900}{2500} = 0.36$$

$$m = X + 1 = 3 + 1 = 4$$

$$\pi_m = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^m = \frac{1 - 0.36}{1 - 0.36^5} \times 0.36^4 = 0.0108$$

$$L_Q = \frac{1 - 0.36}{1 - 0.36^5} \times (0.36^2 + 2 \times 0.36^3 + 3 \times 0.36^4) = 0.176 \text{ packets}$$

$$W = \frac{L_Q}{\lambda(1 - \pi_m)} + \frac{1}{\mu} + \delta = \frac{0.176}{900 \times (1 - 0.0108)} + \frac{1}{25000} + 10^{-5} = 247.7 \mu\text{sec}$$

$$\pi_m = \frac{1 - \rho}{1 - \rho^{m+1}} \times \rho^m$$

$$L_Q = \sum_{n=2}^m (n - 1) \times \pi_n$$

$$W_Q = \frac{L_Q}{\lambda(1 - \pi_m)}$$

$$W = W_Q + \frac{1}{\mu} + \delta$$